

A full solution of Rubin's nonconvex axial lower-dimensional Busemann–Petty conjecture

Candidate solution packet for arXiv:math/0701317

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Status. Candidate full solution, likely valid, needs expert review. The novelty search is bounded and originality remains provisional.

Abstract

Rubin asked whether convexity is essential in the positive part of his lower-dimensional Busemann–Petty theorem for bodies with generalized axial symmetry. We prove the conjectured negative statement for every admissible parameter. The construction starts from an explicit smooth K_ℓ -symmetric star body B whose $(n - i)$ th radial power is a constant plus a degree-two spherical harmonic. The inverse generalized cosine transform multiplies that harmonic by $-i/(n - i)$, producing a negative representing density. Rubin's Radon–cosine perturbation then gives a nearby star body A with all i -dimensional central sections no larger than those of B , but with larger n -volume. The same parameter inequality gives strictly negative curvature in a coordinate two-plane, and this persists under the perturbation, so A is nonconvex as required.

1 Source conjecture and scope

The source is Boris Rubin, *The Lower Dimensional Busemann–Petty Problem for Bodies with the Generalized Axial Symmetry*, arXiv:math/0701317v2 (2007), later published in the *Israel Journal of Mathematics* [1]. The conjecture appears on source PDF page 4, after Theorem 1.1.

The validity of the implication (1.1) $\rightarrow$ (1.2) in (i) and (ii) is an open problem. After several attempts to attack it, we have got an impression that the difficulties here have the same nature as those in the original LDBP for $i = 2$ and 3.

³⁰. Another intriguing open problem is to check the following
Conjecture. *In the case (b) of Theorem 1.1, i.e., when $i = \ell + 1$ or $i = \ell + 2$, there exist a non-convex K_ℓ -symmetric body A and an o.s. star body B so that*

$$\begin{aligned} \text{vol}_i(A \cap \xi) &\leq \text{vol}_i(B \cap \xi) \quad \text{for all } \xi \in G_{n,i} \text{ (not only for } \xi \in G_{n,i}^\ell), \\ \text{but } \text{vol}_n(A) &> \text{vol}_n(B); \text{ cf. [G, Theorem 8.2.4] for } n = 3, \ell = 1. \end{aligned}$$

Figure 1: The full source conjecture on PDF page 4.

In paraphrase, Rubin fixes the parameters from Theorem 1.1(b), namely

$$1 \leq \ell \leq n/2, \quad i + \ell \leq n, \quad i = \ell + 1 \text{ or } i = \ell + 2, \tag{1}$$

and asks for a nonconvex K_ℓ -symmetric body A and an origin-symmetric star body B such that

$$\text{vol}_i(A \cap \xi) \leq \text{vol}_i(B \cap \xi) \quad (\xi \in G_{n,i}), \quad \text{vol}_n(A) > \text{vol}_n(B). \tag{2}$$

Here $K_\ell = O(n - \ell) \times O(\ell)$ acts block-diagonally on $\mathbb{R}^{n-\ell} \oplus \mathbb{R}^\ell$.

2 Definitions and source identities

Write $k = n - i$. For an origin-symmetric star body K , let ρ_K denote its radial function. In Rubin's normalization, the generalized cosine transform is

$$(M^\alpha f)(u) = \gamma_n(\alpha) \int_{S^{n-1}} f(\theta) |\theta \cdot u|^{\alpha-1} d\theta,$$

initially where the integral converges and then by meromorphic continuation. The parameters used below are not poles. On even smooth functions, Rubin's Lemma 4.1 gives

$$M^\alpha M^\beta f = f \quad \text{when} \quad \alpha + \beta = 2 - n. \quad (3)$$

The transform is self-adjoint. Rubin's Lemma 4.2 gives, for a positive constant $c_{n,i}$,

$$R_i M^{1-i} h(\xi) = c_{n,i} R_k h(\xi^\perp), \quad \xi \in G_{n,i}. \quad (4)$$

An origin-symmetric star body K is a k -intersection body precisely when there is a nonnegative finite even Borel measure μ on S^{n-1} such that

$$\rho_K^k = M^{1-k} \mu \quad (5)$$

in the distributional sense [1, Definition 4.5].

We use one elementary Funk–Hecke calculation. If Y is a spherical harmonic of degree two, then

$$M^{1-i} 1 = c_0, \quad M^{1-i} Y = -c_0 \frac{i}{k} Y, \quad c_0 = \sigma_{n-1} \frac{\Gamma(i/2)}{\Gamma(k/2)} > 0. \quad (6)$$

Indeed, for general α , the degree-two multiplier divided by the constant multiplier is

$$-\frac{1 - \alpha}{n - 1 + \alpha};$$

substitution of $\alpha = 1 - i$ gives $-i/k$.

3 Idea of the proof

The positive theorem in the source is powered by positivity of an intersection-body representing density. We force exactly that density to change sign with the lowest nonconstant harmonic compatible with K_ℓ -symmetry. Put

$$Y(\theta) = |\theta''|^2 - \frac{\ell}{n}, \quad \theta = (\theta', \theta'') \in \mathbb{R}^{n-\ell} \oplus \mathbb{R}^\ell.$$

This is an even K_ℓ -invariant degree-two harmonic. If $\rho_B^k = 1 + \varepsilon Y$, formula (6) says that the representing density is proportional to $1 - \varepsilon(i/k)Y$. There is a nonempty interval of ε for which the radial function stays positive but this density is negative near the ℓ -block axis.

Once this sign change is available, a nonnegative bump h supported in the negative region feeds directly into Rubin's perturbation $\rho_A^i = \rho_B^i - \delta M^{1-i} h$. Identity (4) gives the section inequalities on the entire Grassmannian, while self-adjointness and (3) make the mixed radial integral increase. Holder's inequality then forces the strict volume reversal. A short polar-curvature calculation shows that B is already strictly nonconvex, and an arbitrarily small perturbation preserves this strict defect for A .

4 The counterexamples

Theorem 1. *Assume (1). Then there are infinitely smooth, origin-symmetric, K_ℓ -symmetric star bodies $A, B \subset \mathbb{R}^n$ such that A and B are nonconvex and (2) holds. In particular, Rubin's conjecture is true for every parameter choice in its stated scope.*

Proof. Set $k = n - i$. The assumptions give $k \geq \ell \geq 1$. On S^{n-1} write

$$t = |\theta''|^2, \quad Y(\theta) = t - \frac{\ell}{n}.$$

The polynomial $|x''|^2 - (\ell/n)|x|^2$ is harmonic, so Y is an even degree-two spherical harmonic. It is also K_ℓ -invariant, and its range is

$$-\frac{\ell}{n} \leq Y \leq 1 - \frac{\ell}{n}.$$

Choose

$$\frac{kn}{i(n-\ell)} < \varepsilon < \frac{n}{\ell}. \quad (7)$$

The interval is nonempty: after canceling positive factors, its nonemptiness is exactly $k\ell < i(n-\ell)$, which reduces to $\ell < i$. Define B by

$$\rho_B(\theta)^k = 1 + \varepsilon Y(\theta). \quad (8)$$

The upper bound in (7) makes the right-hand side strictly positive. Thus B is an infinitely smooth, origin-symmetric, K_ℓ -symmetric star body.

Let

$$\varphi = M^{1-i}(\rho_B^k).$$

By (6),

$$\varphi(\theta) = c_0 \left[1 - \varepsilon \frac{i}{k} \left(t - \frac{\ell}{n} \right) \right]. \quad (9)$$

At $t = 1$, the lower bound in (7) gives

$$\varphi = c_0 \left[1 - \varepsilon \frac{i(n-\ell)}{kn} \right] < 0.$$

Consequently, φ is negative on a nonempty open K_ℓ -invariant subset of the sphere.

Since $(1-k) + (1-i) = 2-n$, identity (3) gives

$$\rho_B^k = M^{1-k}\varphi.$$

This representing distribution is unique. Indeed, on the even spherical harmonic of degree $2j$, the multiplier of M^{1-k} is a nonzero constant times

$$(-1)^j \frac{\Gamma(j+k/2)}{\Gamma(j+i/2)},$$

so M^{1-k} is injective on even distributions. Therefore no nonnegative measure can represent (8), because the unique representing measure has the smooth density (9), which is negative on an open set. Thus B is not a k -intersection body.

We also need a strict, stable proof of nonconvexity. Choose unit vectors $u \in \mathbb{R}^{n-\ell}$ and $v \in \mathbb{R}^\ell$, and examine the radial curve of $B \cap \text{span}\{u, v\}$. With ω measured from u toward v , it is

$$r(\omega) = (a + \varepsilon \sin^2 \omega)^{1/k}, \quad a = 1 - \frac{\varepsilon \ell}{n} > 0.$$

For a smooth polar curve, the numerator determining signed curvature is $r^2 + 2(r')^2 - rr''$. At $\omega = 0$ this equals

$$a^{2/k-1} \left(a - \frac{2\varepsilon}{k} \right). \quad (10)$$

The last factor is negative exactly when

$$\varepsilon > \frac{nk}{k\ell + 2n}.$$

Put $s = i - \ell \in \{1, 2\}$. Since $k = n - \ell - s$,

$$(k\ell + 2n) - i(n - \ell) = n(2 - s) \geq 0.$$

Hence the lower bound in (7) is at least the curvature threshold, and the inequality is strict. Expression (10) is negative, so this planar section is not locally convex. Therefore B is nonconvex.

Let $\Omega = \{\theta : \varphi(\theta) < 0\}$. Choose a nonzero smooth, even, K_ℓ -invariant function $h \geq 0$ supported in Ω . For $\delta > 0$, define

$$\rho_A^i = \rho_B^i - \delta M^{1-i} h. \quad (11)$$

For all sufficiently small δ , the right-hand side is positive, so it defines an infinitely smooth, origin-symmetric, K_ℓ -symmetric star body A . Moreover, $\rho_A \rightarrow \rho_B$ in C^∞ as $\delta \rightarrow 0$. The curvature in (10) is strictly negative, so it remains negative for all sufficiently small δ . Thus A is nonconvex.

For every $\xi \in G_{n,i}$, (4) yields

$$R_i \rho_A^i(\xi) = R_i \rho_B^i(\xi) - \delta c_{n,i} R_k h(\xi^\perp) \leq R_i \rho_B^i(\xi).$$

Since $\text{vol}_i(K \cap \xi) = i^{-1} R_i \rho_K^i(\xi)$, this is the first inequality in (2) on the full Grassmannian.

Finally, self-adjointness of the cosine transform and (3) give

$$\begin{aligned} \langle \rho_B^k, \rho_B^i - \rho_A^i \rangle &= \delta \langle M^{1-k} \varphi, M^{1-i} h \rangle \\ &= \delta \langle \varphi, M^{1-k} M^{1-i} h \rangle \\ &= \delta \langle \varphi, h \rangle < 0. \end{aligned}$$

Thus

$$\int_{S^{n-1}} \rho_B^n < \int_{S^{n-1}} \rho_B^k \rho_A^i.$$

Holder's inequality, with exponents n/k and n/i , gives

$$\int_{S^{n-1}} \rho_B^k \rho_A^i \leq \left(\int_{S^{n-1}} \rho_B^n \right)^{k/n} \left(\int_{S^{n-1}} \rho_A^n \right)^{i/n}.$$

It follows that $\int \rho_B^n < \int \rho_A^n$, and the polar volume formula proves $\text{vol}_n(B) < \text{vol}_n(A)$. This completes the construction. $\square$

Remark 1 (What is explicit). *The base body B is given explicitly by (8), with any ε in (7). The body A is an existence-level smooth perturbation: any nonzero invariant bump supported where (9) is negative works once δ is sufficiently small. No unproved analytic or numerical dependency remains.*

5 Verification

The accompanying script `code/verifier.py` enumerates every admissible triple (n, ℓ, i) with $3 \leq n \leq 100$, chooses the midpoint of (7), and checks positivity of ρ_B^k , negativity of (9) at $t = 1$, negativity of the curvature factor in (10), and a 201-point angular sampling of the radial-power profile. All tested cases pass. This code is only an algebraic sanity check; the proof uses the exact inequalities and transform identities above.

The main expert-review points are:

1. the normalization and degree-two multiplier in (6);
2. injectivity of M^{1-k} on even distributions at the stated integer parameters;
3. the index substitution in Rubin’s inversion and intertwining lemmas;
4. persistence of the strict curvature defect under (11).

6 Novelty search and limitations

A bounded search on 11 August 2026 covered the run’s registry, solution, attempt, and proof-gap indexes; the exact arXiv id and title; phrases from the source conjecture; the published DOI; and later work on intersection bodies with symmetries, including arXiv:1307.3206. The existing run packet for arXiv:0704.0061 concerns a different (q, ℓ) -ball λ -intersection-body question and merely lists the present source as related literature. No explicit later resolution or this degree-two construction was found. The search is not exhaustive, so originality is provisional.

This theorem addresses the conjecture after Theorem 1.1(b), under the source’s condition $i + \ell \leq n$. It does not address the separate convex exceptional cases with $i + \ell > n$ discussed immediately before the conjecture; those remain tied to the open two- and three-dimensional-section problem.

References

- [1] B. Rubin, *The Lower Dimensional Busemann-Petty Problem for Bodies with the Generalized Axial Symmetry*, arXiv:math/0701317v2 (2007), *Israel J. Math.* **173** (2009), 213–233. <https://arxiv.org/abs/math/0701317>.